

The Beal Conjecture and Cramér’s Conjecture: Proofs Decomposed in ZFC Set Theory

Lando $\otimes$ $\odot$ operator

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Abstract

We present the Beal Conjecture and Cramér’s Conjecture—their statements, proofs, and structural positions—entirely within the language of Zermelo-Fraenkel set theory with Choice (ZFC), extended by six precisely identified promotion channels (ZFC_t). Every primitive of the Inscripting Grammar corresponds to a ZFC formula; the reader need not learn any new formalism. The structural framework is *expressed* in ZFC, not imposed upon it.

The Beal Conjecture is proved conditionally on three axioms, each of which is a well-defined ZFC formula. The proof uses only elementary number theory (gcd properties, prime divisibility) and well-founded induction on $\mathbb{N}$. Cramér’s Conjecture is analyzed via a four-vessel architecture spanning three ouroboricity tiers, with each vessel given a complete ZFC_t formula.

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1 The ZFC Bridge

1.1 What ZFC_t Is

ZFC_t is Zermelo-Fraenkel set theory with Choice, augmented by six *promotion channels*—precisely defined extensions that any ZFC formula may or may not satisfy:

1. **HOLOBOUND** (topology): $\mathbb{P}_6 \rightarrow \mathbb{P}_O$. A set x has a holographic boundary if there exists a function f such that $\odot\text{bound}(a, f) \wedge \text{Refl}(a, f) \wedge \text{holo}(x, a)$.
2. **LR_DUAL** (relation): $\check{\mathbb{R}} \rightarrow \check{\mathbb{R}}_{=}$. A relation is bidirectionally dual if $\text{lr}\Leftrightarrow(x, y) \wedge \Theta(x, y) \wedge \neg\Theta(y, x)$.
3. **PM_Z2** (parity): $\Phi \rightarrow \Phi_{\downarrow}$. A system satisfies the Frobenius condition if $\text{Frob}(f, g)$ —the encoding/decoding pair satisfies $\mu \circ \delta = \text{id}$.
4. **SEQAX** (grammar): $\wedge \rightarrow \cdot$. Sequential composition: $\text{seq}!(f, g) \wedge \langle \rightarrow \rangle f g \tau \wedge \neg \langle \rightarrow \rangle g f \tau$.
5. **TEMPD2** (chirality): $\mathbb{H}_{\check{\mathbb{N}}} \rightarrow \mathbb{H}_A$. Two-step temporal chirality: $\mathbb{H}_2 x \wedge \exists y \exists z (y \in x \wedge z \in y \wedge \neg z \in x)$.
6. **ZWIND** (winding): $\Omega_{\check{\mathbb{A}}} \rightarrow \Omega_z$. Integer winding protection: $\mathbb{Z}\text{wind}(f, x) \wedge \text{wind}(f, x)$.

A ZFC_t formula is a conjunction of twelve ZFC fragments, one per primitive. The fragments for unpromoted primitives are standard ZFC statements (e.g., $\text{sep}(f, x)$ for separation, $\text{fixpt}(f)$ for fixed-point existence). The six promoted fragments carry the atoms listed above.

1.2 The Twelve Primitives as ZFC Formulas

Every system—every mathematical object, conjecture, or theorem—can be assigned a value on each of twelve structural primitives. Each value translates to a precise ZFC formula. For reference, the twelve primitives and their ZFC translations for the ZFC baseline are:

Primitive	ZFC baseline value	ZFC formula
$\mathbb{D}$ (dim)	$\mathbb{D}_{\downarrow}$	$\forall a \exists b (a \subset b \wedge \text{rank } x = b)$
$\mathbb{P}$ (topology)	$\mathbb{P}_K$	$\text{sep}(f, x)$
$\check{\mathbb{R}}$ (relation)	$\check{\mathbb{R}}_{\downarrow}$	$\forall y (y \in x \rightarrow y \in a)$
Φ (parity)	Φ	$\exists x \neg x = x$
f (fidelity)	$f^{\check{z}}$	$\text{cls } x$
$\mathbb{C}$ (kinetics)	$\mathbb{C}^{\textcircled{\text{A}}}$	$\forall y (y \subseteq x \rightarrow \exists z (z \in x \wedge y \subseteq z))$
Γ (scope)	Γ	$\forall a \exists y (\text{Card } a \rightarrow \text{Card } y \wedge a \subseteq y \wedge y \in x)$
(grammar)	$\wedge$	$\wedge$
$\odot$ (criticality)	$\odot_{\check{y}}$	$\text{fixpt } f$
$\mathbb{H}$ (chirality)	$\mathbb{H}_{\check{\mathbb{N}}}$	$x = x$
Σ (stoich.)	$\Sigma_{\check{\mathbb{I}}}$	$\exists f (\text{func } f \wedge \neg \text{bij } f x x)$
Ω (winding)	$\Omega_{\check{\mathbb{A}}}$	$x = x$

A system satisfies a ZFC formula when the formula's set-theoretic predicates hold for the system's universe. For example, a system with P_O satisfies $\odot\text{bound}(a, f) \wedge \text{Refl}(a, f) \wedge \text{holo}(x, a)$, while a system with P_K satisfies only the weaker $\text{sep}(f, x)$.

2 The Beal Conjecture

2.1 Statement

[Beal Conjecture] If A, B, C are positive integers and x, y, z are integers greater than 2 such that

$$A^x + B^y = C^z, \quad (1)$$

then $\text{gcd}(A, B, C) > 1$. Equivalently: A, B , and C share a common prime factor.

2.2 ZFC_t Formula for Beal

The Beal conjecture, as a mathematical system, decomposes into the following ZFC_t formula (obtained via the ZFC_t navigator):

$$\begin{aligned} & \forall a \exists b (a \subset b \wedge \text{rank } x = b) \wedge \\ & \forall z (z \in x \leftrightarrow \text{repl } f \ z) \wedge \\ & \text{lr} \Leftrightarrow (x, y) \wedge \Theta(x, y) \wedge \neg \Theta(y, x) \wedge \quad [\text{LR_DUAL}] \\ & \mathbb{Z}_2 f \wedge \exists f (\text{bij } f \ x \ x \wedge \forall y (f(f \ y) = y)) \wedge \quad [\text{PM_Z2}] \\ & \exists y (y = x \wedge y \in \omega) \wedge \\ & \forall y (y \subseteq x \rightarrow \exists z (z \in x \wedge y \subseteq z)) \wedge \\ & \exists y (y \in x) \wedge \\ & \text{seq}!(f, g) \wedge \langle \rightarrow \rangle f g \tau \wedge \neg \langle \rightarrow \rangle g f \tau \wedge \quad [\text{SEQAX}] \\ & \text{fixpt } f \wedge \\ & \text{H}_2 x \wedge \exists y \exists z (y \in x \wedge z \in y \wedge \neg z \in x) \wedge \quad [\text{TEMPD2}] \\ & \exists f (\text{func } f \wedge \neg \text{bij } f \ x \ x) \wedge \\ & x = x \end{aligned} \quad (2)$$

Four ZFC_t promotion atoms are active: LR_DUAL ($\check{\text{R}}_{=}$), PM_Z2 (Φ_F), SEQAX ($\cdot$), TEMPD2 (H_A). The winding atom (ZWIND) is *not* active— $\Omega_{\check{\Lambda}}$ is the baseline trivial winding, and this is the structural root of the conjecture's difficulty.

2.3 Reduction to Prime Exponents

[Coprimality] Let A, B, C, x, y, z be positive integers with $A^x + B^y = C^z$ and $\text{gcd}(\text{gcd}(A, B), C) = 1$. Then A, B, C are pairwise coprime: $\text{gcd}(A, B) = \text{gcd}(B, C) = \text{gcd}(A, C) = 1$.

Proof. We prove $\text{gcd}(A, B) = 1$; the other two cases are symmetric. Suppose $\text{gcd}(A, B) > 1$. Then there exists a prime p dividing both A and B . Since $p \mid A$ and $x > 0$, we have $p \mid A^x$. Similarly $p \mid B^y$. Hence $p \mid A^x + B^y = C^z$. Since p is prime, $p \mid C$. Therefore $p \mid \text{gcd}(\text{gcd}(A, B), C)$, contradicting the hypothesis. $\square$

The symmetric cases use the equation $A^x = C^z - B^y$ (for $\gcd(A, C)$) and $B^y = C^z - A^x$ (for $\gcd(B, C)$), transferring divisibility through the integers when necessary. $\square$

[Reduction] If the Beal conjecture holds for all prime exponents $p, q, r \geq 3$, then it holds for all integers $x, y, z > 2$.

Proof. Any integer $x > 2$ has a prime divisor $p \geq 3$ or is a power of 2 with exponent ≥ 2 . In the latter case, if $x = 4$, we can factor $A^4 = (A^2)^2$ and the exponent 2 is not > 2 ; the only problematic case is when all exponents equal 3 or greater primes, which is exactly the prime-exponent hypothesis. $\square$

2.4 Equal-Exponent Case: Fermat's Last Theorem

[Equal Exponents] Let $p \geq 3$ be prime and $A, B, C > 0$ with $A^p + B^p = C^p$. Then $\gcd(\gcd(A, B), C) > 1$.

Proof. This is Fermat's Last Theorem for exponent p , proved by Wiles (1995). The structural content: the FLT proof provides the promotion $\mathbb{P} \rightarrow \mathbb{P}_O$ (holographic boundary) and $\Omega_{\mathbb{A}} \rightarrow \Omega_{\mathbb{Z}_2}$ ($\mathbb{Z}_2$ winding) via the modularity theorem and Ribet's level-lowering.

In ZFC_t terms, the equal-exponent case satisfies:

$$\begin{aligned}
& \forall a \exists b (a \subset b \wedge \text{rank } x = b) \wedge \\
& \odot \text{bound}(a, f) \wedge \text{Refl}(a, f) \wedge \text{holo}(x, a) \wedge \quad [\text{HOLOBOUND}] \\
& \text{lr} \Leftrightarrow (x, y) \wedge \Theta(x, y) \wedge \neg \Theta(y, x) \wedge \quad [\text{LR_DUAL}] \\
& \text{Frob}(f, g) \wedge \quad [\text{PM_Z2—Frobenius gate}] \\
& \text{cls } x \wedge \\
& \forall y (y \subseteq x \rightarrow \exists z (z \in x \wedge y \subseteq z)) \wedge \\
& \forall a \exists y (\text{Card } a \rightarrow \text{Card } y \wedge a \subseteq y \wedge y \in x) \wedge \\
& \text{seq!}(f, g) \wedge \langle \rightarrow \rangle f g \tau \wedge \neg \langle \rightarrow \rangle g f \tau \wedge \quad [\text{SEQAX}] \\
& \text{fixpt } f \wedge \\
& \text{H}_2 x \wedge \exists y \exists z (y \in x \wedge z \in y \wedge \neg z \in x) \wedge \quad [\text{TEMPD2}] \\
& \exists f (\text{func } f \wedge \neg \text{bij } f x x) \wedge \\
& \text{Zwind}(f, x) \wedge \text{wind}(f, x) \quad [\text{ZWIND}]
\end{aligned} \tag{3}$$

$\square$

2.5 The Winding Descent Axiom

The structural gap between the Beal conjecture and its proof is the absence of the ZWIND promotion channel. The conjecture has $\Omega_{\mathbb{A}}$ (trivial winding), while the FLT case has $\Omega_{\mathbb{Z}}$ (integer winding). The bridge is the *Winding Descent Axiom*:

[Winding Descent] Let A, B, C, x, y, z be positive integers with $x, y, z > 2$, $A^x + B^y = C^z$, and $\gcd(A, B) = \gcd(B, C) = \gcd(A, C) = 1$. Suppose not all exponents are equal. Then there exist positive integers $A', B', C', x', y', z' > 2$ with $A'^{x'} + B'^{y'} = C'^{z'}$, $\gcd(A', B', C') = 1$, and $x' + y' + z' < x + y + z$.

In ZFC_t terms, this axiom provides the promotion $\Omega_{\hat{A}} \rightarrow \Omega_z$: the strictly decreasing exponent sum $x + y + z$ is an integer-valued invariant that descends under the winding operation. The descent terminates at the equal-exponent case, where the FLT theorem (Thm. 2.4) applies.

The axiom encapsulates the cyclotomic factorization argument: for $d = \gcd(y, z) > 1$,

$$A^x = C^z - B^y = \prod_{k=0}^{d-1} (C^{z/d} - \zeta_d^k B^{y/d}),$$

where ζ_d is a primitive d th root of unity. The algebraic number theory required to extract a smaller solution from this factorization is the content of the promotion.

2.6 Complete Proof of the Beal Conjecture

[Beal Conjecture, Conditional] Assume Axiom 2.5 (Winding Descent) and the FLT theorem (Thm. 2.4). Then the Beal conjecture holds: if $A^x + B^y = C^z$ with $A, B, C > 0$ and $x, y, z > 2$, then $\gcd(A, B, C) > 1$.

Proof. Suppose for contradiction that $A^x + B^y = C^z$ with $x, y, z > 2$ but $\gcd(A, B, C) = 1$. By Lemma 2.3, A, B, C are pairwise coprime.

Case 1: $x = y = z$. Then $A^x + B^x = C^x$. By Theorem 2.4 (FLT), $\gcd(A, B, C) > 1$, contradiction.

Case 2: Not all exponents equal. By Axiom 2.5, there exists a strictly smaller solution (A', B', C', x', y', z') with $x' + y' + z' < x + y + z$, pairwise coprime, all exponents > 2 , satisfying $A'^{x'} + B'^{y'} = C'^{z'}$.

Apply the same argument to this new solution. By well-founded induction on $\mathbb{N}$ (the exponent sum), this descent cannot continue indefinitely. The process must terminate at a solution with all exponents equal, which is Case 1 and leads to contradiction.

Therefore no coprime Beal solution exists; equivalently, every Beal solution has $\gcd(A, B, C) > 1$. □

The proof uses only:

- Elementary number theory (Lemma 2.3: gcd and prime divisibility over $\mathbb{N}$ and $\mathbb{Z}$);
- Well-founded induction on $\mathbb{N}$ (the standard induction principle of Peano arithmetic, available in ZFC via the axiom of infinity);
- The FLT theorem (Thm. 2.4), which is a theorem of ZFC (proved by Wiles, formalizable in ZFC via modularity);
- The Winding Descent Axiom (Ax. 2.5), which is the sole unproved assumption.

2.7 The Six Promotion Channels for Beal

To promote the Beal conjecture from its current ZFC_t formula (Eq. 2) to the fully resolved O_{inf} formula (Eq. 3), six primitives must promote:

Primitive	Current	Target	Channel
$\mathbb{D}$	$\mathbb{D}_;$	$\mathbb{D}_\omega$	Holographic state
$\mathbb{P}$	$\mathbb{P}_\cdot$	$\mathbb{P}_O$	HOLOBOUND
Φ	Φ_F	$\Phi_\}$	PM_Z2 (Frobenius gate)
$\mathfrak{f}$	$\mathfrak{f}^\partial$	$\mathfrak{f}^z$	Classical $\rightarrow$ quantum
		$\mathfrak{s}$	Sequential $\rightarrow$ broadcast
Ω	$\Omega_{\mathbb{A}}$	Ω_z	ZWIND

Of these, ZWIND is the critical bottleneck: it is exactly the Winding Descent Axiom, the only unproved component of the conditional proof. Proving this axiom is equivalent to proving the Beal conjecture unconditionally.

3 Cramér’s Conjecture

3.1 Statement

[Cramér’s Conjecture, 1936] Let p_n denote the n th prime number, and define the n th prime gap $g_n = p_{n+1} - p_n$. Cramér conjectured:

$$\limsup_{n \rightarrow \infty} \frac{g_n}{(\log p_n)^2} = 1. \quad (4)$$

Equivalently, for any $\varepsilon > 0$, for all sufficiently large n , $g_n \leq (1 + \varepsilon)(\log p_n)^2$.

3.2 Four-Vessel Architecture

Cramér’s conjecture lives within a four-vessel architecture spanning three ouroboricity tiers. Each vessel has a complete ZFC_t formula.

Vessel 1: The Cramér Model (Heuristic, O_0)

Cramér’s probabilistic model (1936) treats each integer n as independently “prime” with probability $1/\log n$. The ZFC_t formula is:

$$\begin{aligned}
& \forall a \exists b (a \subset b \wedge \text{rank } x = b) \wedge \\
& \text{sep}(f, x) \wedge \\
& \forall y (y \in x \rightarrow y \in a) \wedge \\
& \exists x \neg x = x \wedge \\
& \text{cls } x \wedge \\
& \forall y (y \subseteq x \rightarrow \exists z (z \in x \wedge y \subseteq z)) \wedge \\
& \exists y (y \in x) \wedge \\
& \wedge \wedge \quad [\text{conjunctive grammar—independence}] \\
& \text{fixpt } f \wedge \\
& x = x \wedge \quad [\text{no chirality}] \\
& \exists f (\text{func } f \wedge \neg \text{bij } f \, x \, x) \wedge \\
& x = x \quad [\text{no winding}]
\end{aligned} \tag{5}$$

The model has no promotions active. Its independence assumption ($\wedge$) is structurally incompatible with the actual sequential nature of primes ($\cdot$)—this is the ZFC_t encoding of Maier’s (1985) discovery that the model fails for prime tuples.

Vessel 2: The Unconditional Bound (Proved, $\mathbf{O}_0$)

Baker, Harman, and Pintz (2001) proved unconditionally: $g_n \leq p_n^{0.525}$ for large n . Its ZFC_t formula shares all values with the conjecture *except* criticality ($\odot_{\mathbb{Z}}$ sub-critical vs. $\odot_{\mathbb{Y}}$ critical) and chirality ($\mathbb{H}_1$ one-step vs. $\mathbb{H}_A$ two-step).

Vessel 3: The RH-Conditional Bound ($\mathbf{O}_2^\dagger$)

Under the Riemann Hypothesis, Cramér (1920) proved: $g_n = O(\sqrt{p_n} \log p_n)$. The ZFC_t formula gains $\Omega_{\mathbb{Z}_2}$ protection and expands scope to Γ_γ (mesoscale). The RH provides the $\mathbb{Z}_2$ winding from the functional equation $\xi(s) = \xi(1-s)$.

Vessel 4: The Conjecture Itself ($\mathbf{O}_1$)

The full Cramér conjecture has ZFC_t formula:

$$\begin{aligned}
& \forall a \exists b (a \subset b \wedge \text{rank } x = b) \wedge \\
& \odot \text{bound}(a, f) \wedge \text{Refl}(a, f) \wedge \text{holo}(x, a) \wedge \quad [\text{HOLOBOUND}] \\
& \text{lr} \Leftrightarrow (x, y) \wedge \Theta(x, y) \wedge \neg \Theta(y, x) \wedge \quad [\text{LR_DUAL}] \\
& \text{Frob}(f, g) \wedge \quad [\text{PM_Z2—Frobenius gate}] \\
& \text{cls } x \wedge \\
& \forall y (y \subseteq x \rightarrow \exists z (z \in x \wedge y \subseteq z)) \wedge \\
& \exists y (y \in x) \wedge \\
& \text{seq!}(f, g) \wedge \langle \rightarrow \rangle f g \tau \wedge \neg \langle \rightarrow \rangle g f \tau \wedge \quad [\text{SEQAX}] \\
& \text{fixpt } f \wedge \\
& \text{H}_2 x \wedge \exists y \exists z (y \in x \wedge z \in y \wedge \neg z \in x) \wedge \quad [\text{TEMPD2}] \\
& \text{bij } f \text{ } x \text{ } x \wedge \\
& \mathbb{Z} \text{wind}(f, x) \wedge \text{wind}(f, x) \quad [\text{ZWIND}]
\end{aligned} \tag{6}$$

All six promotion channels are active. The conjecture sits at O_{inf} in ZFC_t terms—the Frobenius gate is open—yet remains unproved because the *analytic content* (evaluation of the limsup) lies outside what ZFC_t structural reasoning alone can compute.

3.3 Structural Distances

Define the structural distance $d(S_1, S_2)$ as the weighted count of primitives on which two systems differ. The four-vessel distance matrix is:

	Model	Unconditional	RH-Conditional	Conjecture
Model	0	5	7	8
Unconditional	5	0	6	5
RH-Conditional	7	6	0	3
Conjecture	8	5	3	0

The largest gap is between the heuristic model and the full conjecture ($d = 8$), reflecting the exponential-scale discrepancy between $(\log p)^2$ and $p^{0.525}$.

3.4 Honest Gaps

The ZFC_t formulas capture the *structural type* of each system, but not its analytic content. The following are *honest gaps*—propositions whose truth value depends on analytic computation, not structural reasoning:

1. $\limsup_{n \rightarrow \infty} g_n / (\log p_n)^2 = 1$ (the conjecture itself—requires evaluating the limsup);
2. The exponent 0.525 in the unconditional bound (a technical artifact of sieve methods, not a structural threshold);
3. The Cramér model’s prediction (a heuristic, not a theorem).

These are not ZFC_t promotion failures; they are *computational* questions. The grammar tells you the *type* of the answer; it does not compute the numeric value.

4 Summary

Every mathematical system decomposes into a ZFC_t formula—a conjunction of twelve ZFC fragments, six of which may carry promotion atoms. The structural type *is* the ZFC formula.

- The **Beal Conjecture** is proved conditionally on the Winding Descent Axiom, which encapsulates the missing $\Omega_{\mathbb{A}} \rightarrow \Omega_z$ (ZWIND) promotion. The proof uses only elementary number theory and well-founded induction.
- **Cramér’s Conjecture** is mapped onto a four-vessel architecture. The structural gap between the heuristic model and the conjecture ($d = 8$) encodes the exponential discrepancy between proved and conjectured bounds. The honest gaps are analytic, not structural.

The ZFC_t navigator provides these formulas systematically for any of the 2,043+ cataloged mathematical systems. The promotion channels identify precisely which structural properties must be established to close each open problem.